SAES2026
14th INTERNATIONAL SYMPOSIUM ON APPLIED ENGINEERING AND SCIENCES

Grokking · Generative model · Flow matching · Transformer

Weight Tying Causes Grokking in Conditional Flow Matching

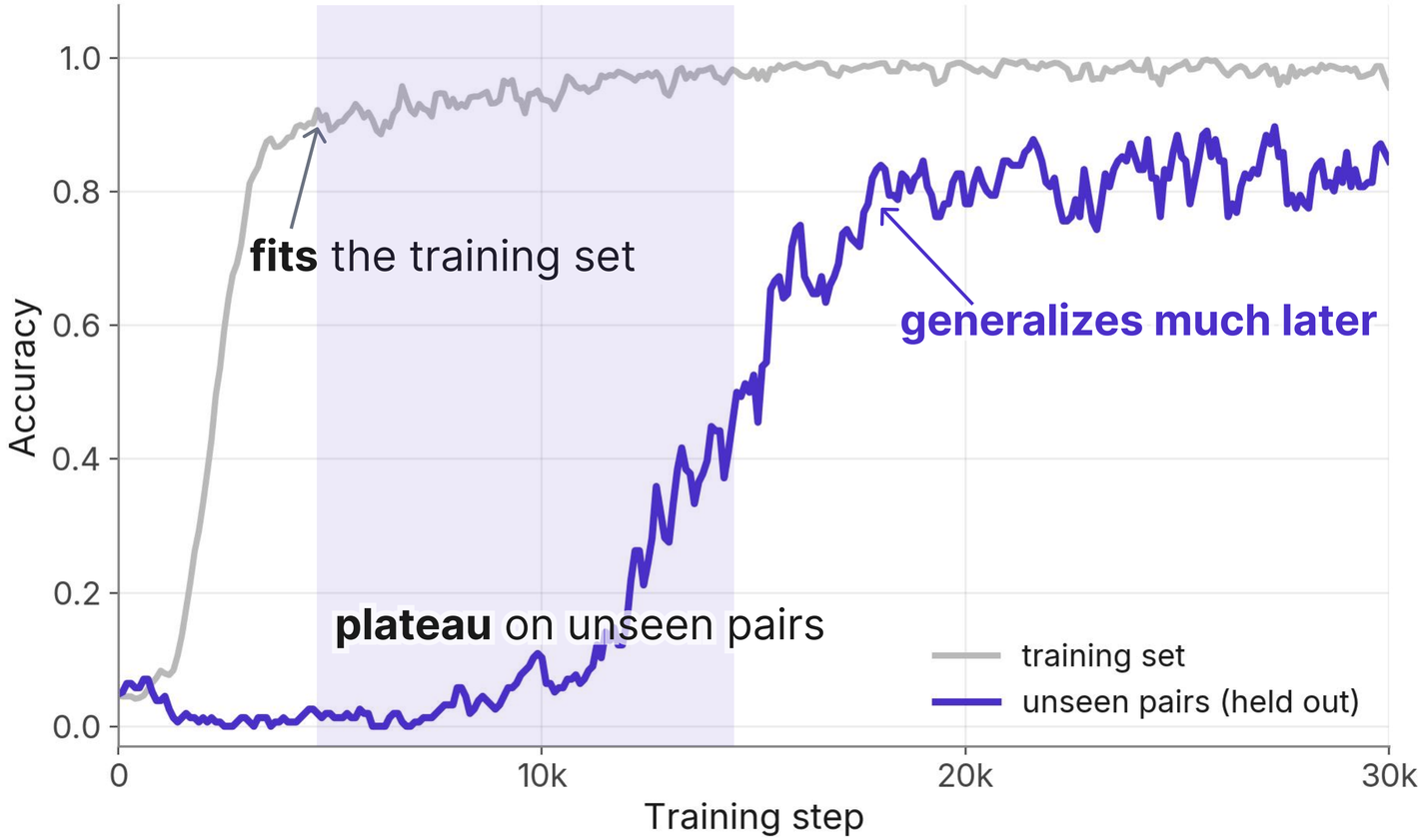
Kosuke Ukita, Tsuyoshi Okita
Kyushu Institute of Technology

■ Background | What is grokking?

What is grokking?

Grokking: [Power+, 2022], [Liu+, 2023], [Kim+, 2026], [Yıldırım, 2026]

Fit first, generalize much later.



Known causes of grokking

- Amount of training data
- Optimization
- Initialization scale
- Normalization / attention

→ **Conditioning-specific factors: unexplored**

■ Research Question

In conditional generators, does the structure of the conditioning mechanism affect grokking?

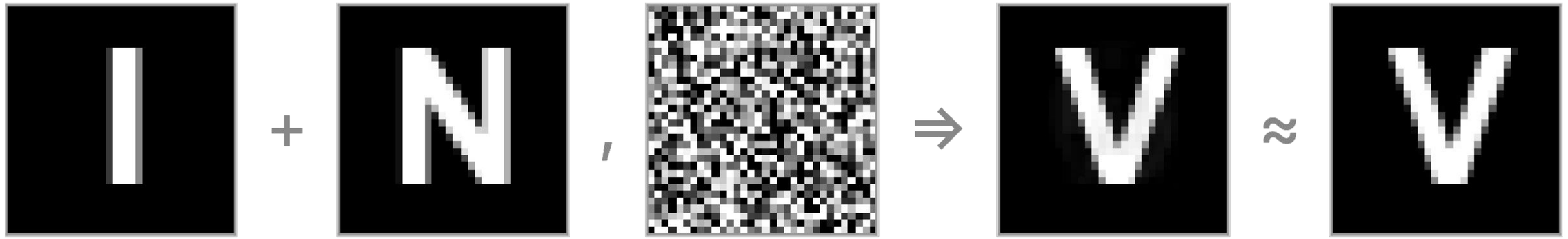
We study one axis of it: whether the condition and the generated target share an input embedding.

■ Approarch

Task: modular addition with images

Generate the image of c = (a + b) mod 23



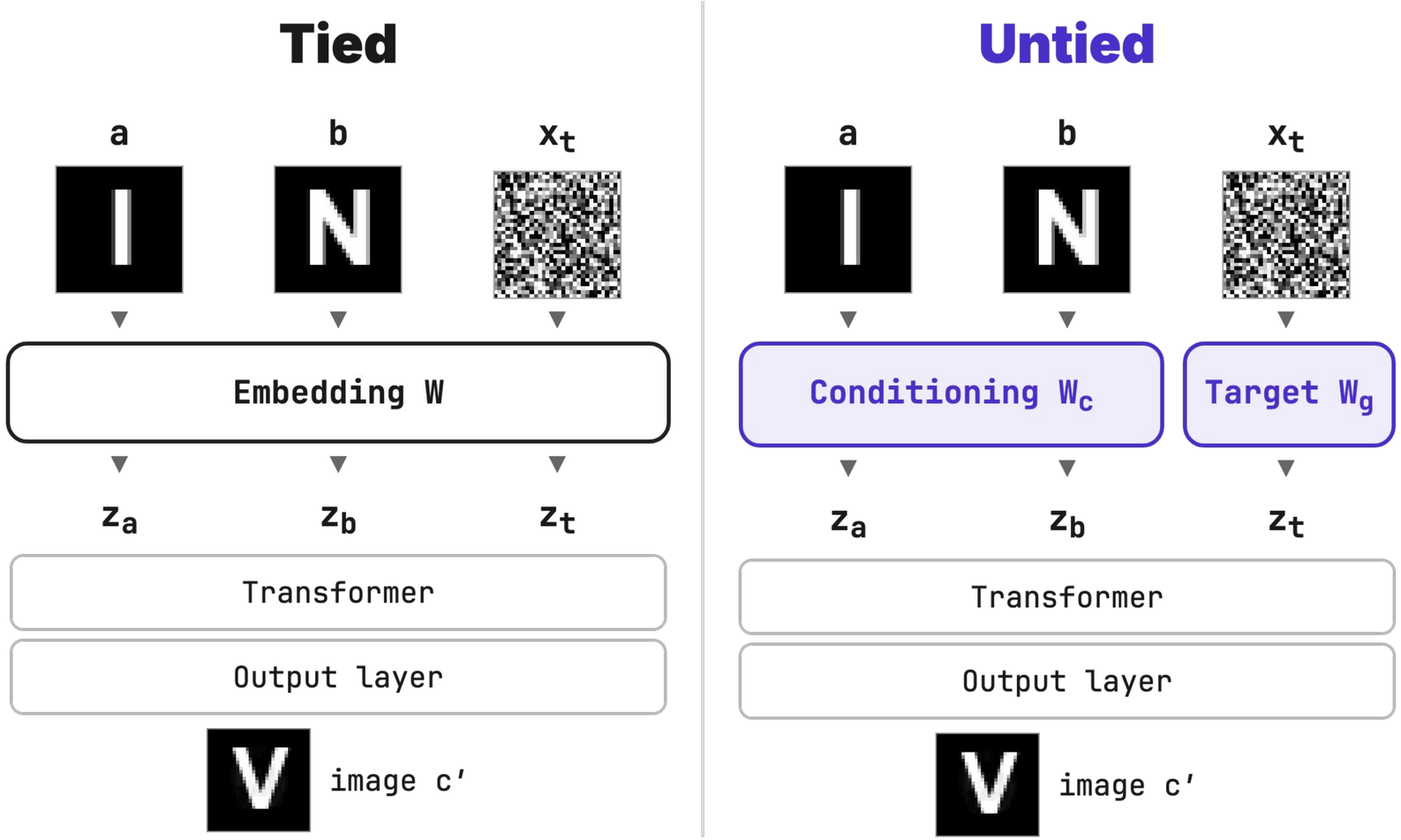


276 pairs: 248 train, 28 held out

Scored only on **unseen pairs**

Also as **16×16×8-frame video**

The only change: tied or untied embedding



Tied (= shared)

One embedding reads the clean conditions **and** the noisy target.

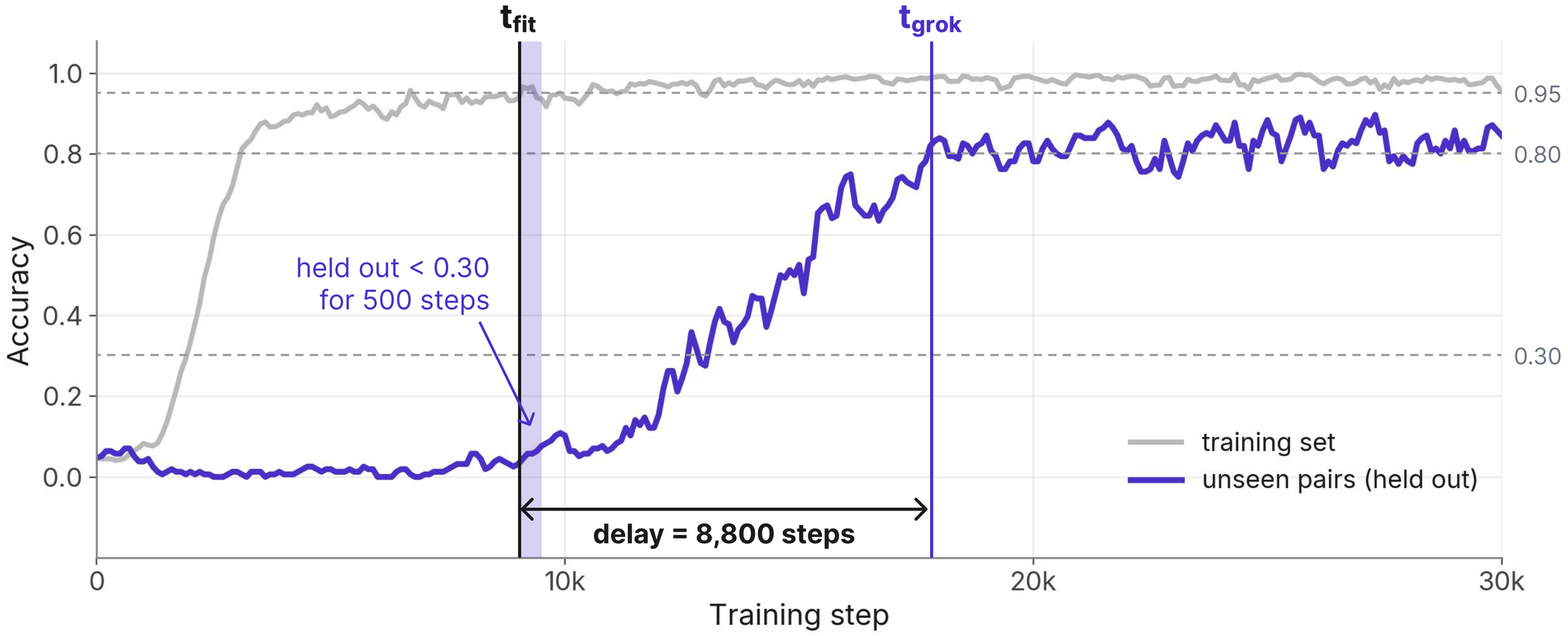
Untied (= unshared)

Separate embeddings for the conditions and for the target.

→ Everything else is identical: 1-layer transformer that generates in one step.

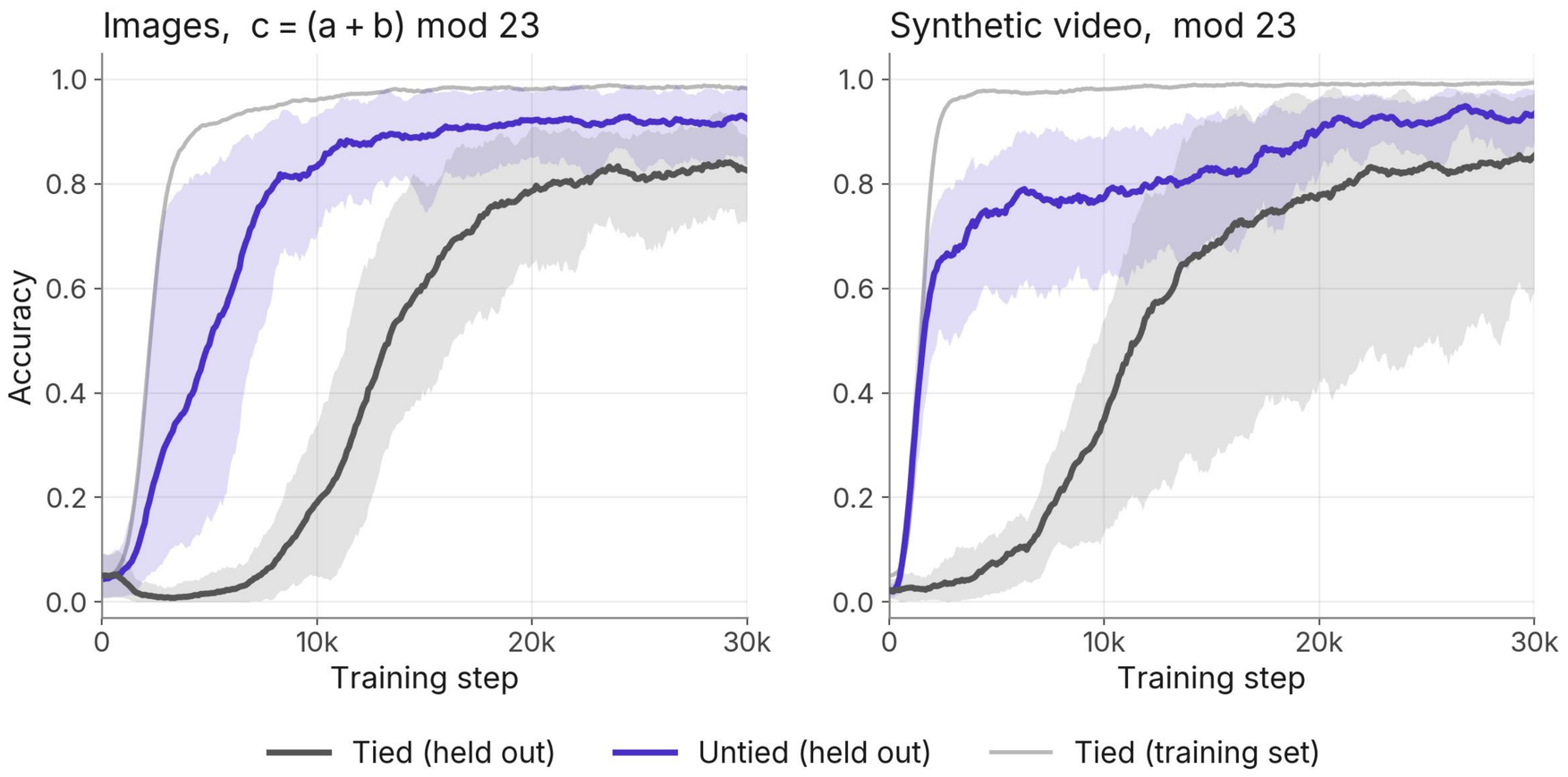
How we detect grokking

A run groks only if all three time points exist



■ Results

Tied groks, Untied mostly does not



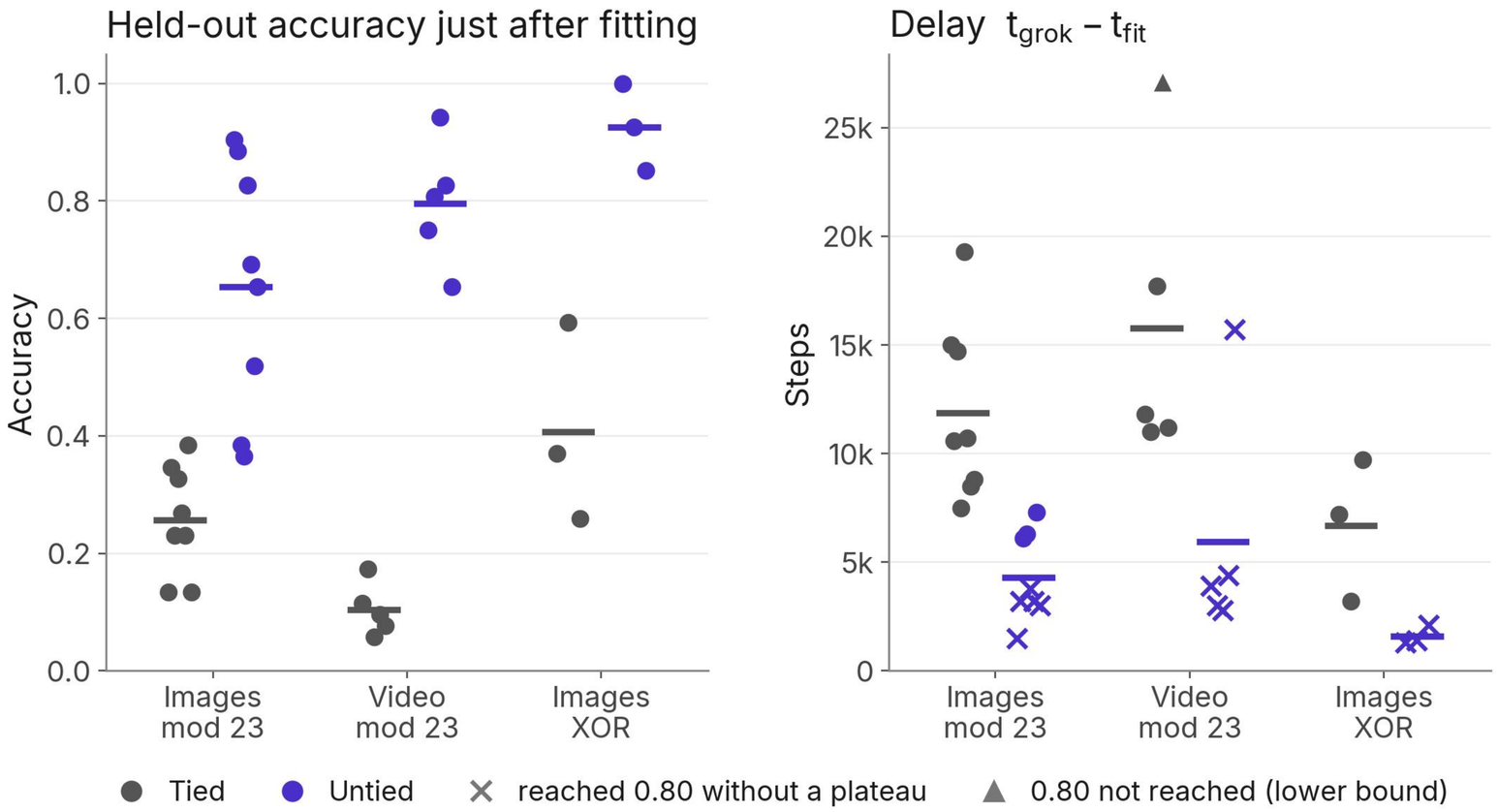
Grokking was observed

Tied 94% of runs

Untied 19% of runs

Untied accuracy rises
together with training accuracy.

Tied has a memorization phase to escape



Held-out accuracy just after fitting

Tied **0.10–0.41** vs Untied **0.65–0.93**

Untied is higher in **96 of 98** tied-vs-untied run pairs.

Delay

2.6–4.2× longer with Tied.

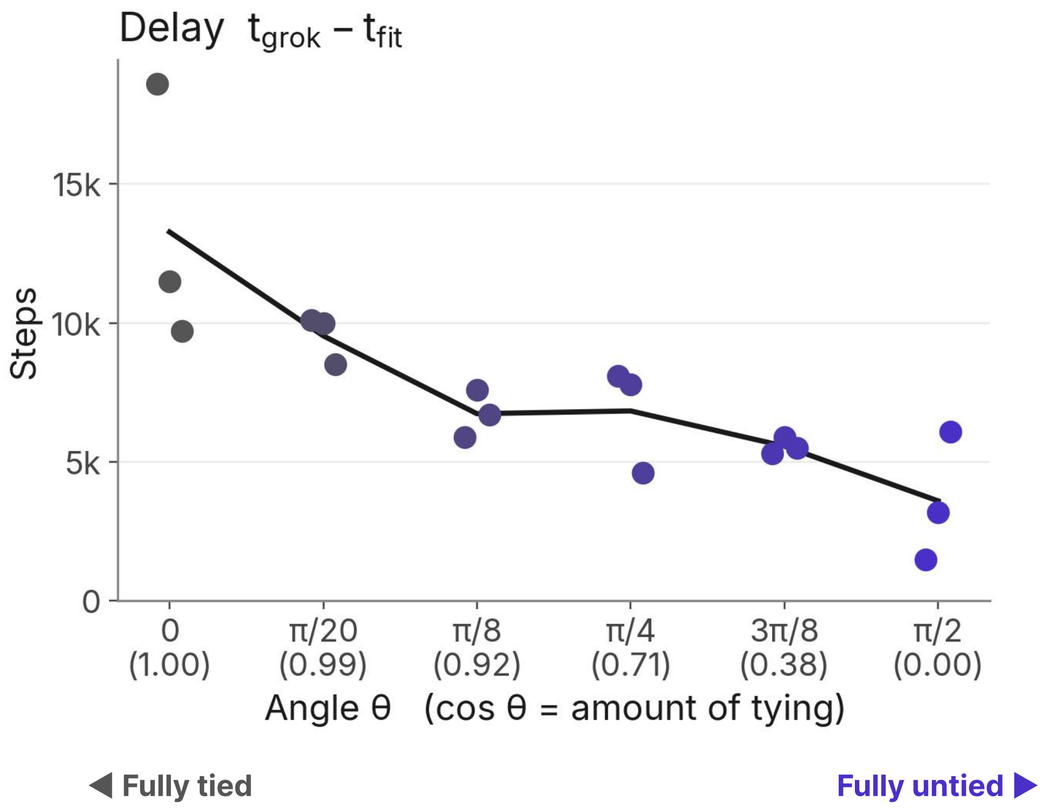
Not the task, not capacity

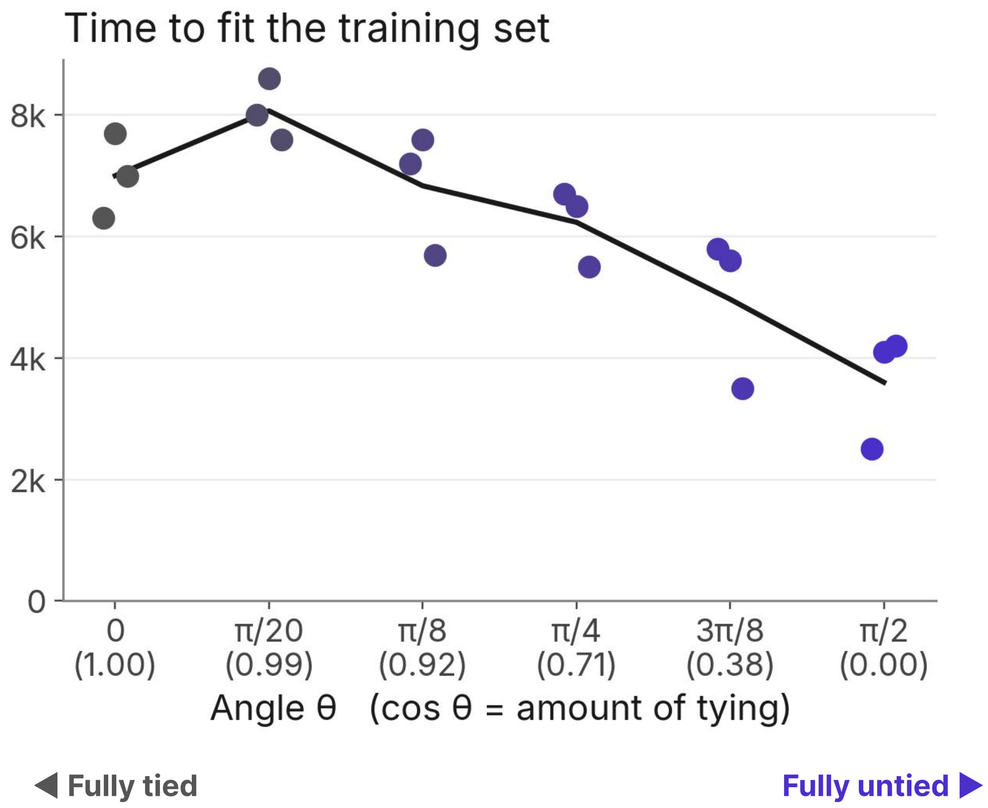
Holds on 2 modalities and 2 groups.

Untied even has **more** parameters.

Tying is a quantity, not a switch

$$W_{\text{target}} = \cos\theta\, W_{\text{shared}} + \sin\theta\, W_{\text{private}}$$

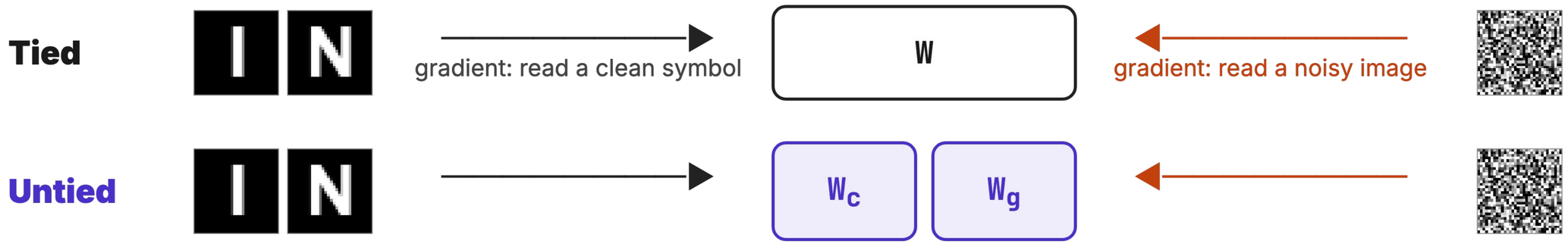




Less tying → shorter delay (Spearman ρ = 0.86, seeds = 18) and **faster fitting**, at an identical parameter count.

■ Discussion

Grokking here is a delay without benefit



Why Tied is slow

One matrix gets **both** gradients every step. The model **memorizes until a representation that serves both roles** emerges.

What untying gives

- **Faster: the delay is 2.6–4.2× shorter**
- **More accurate right after fitting**

Summary

Question	Does the conditioning mechanism's structure affect grokking? (axis: a shared condition–target embedding)
Finding	Tying the condition and target embeddings causes the delay: grokking in 94% of Tied runs vs 19% of Untied.
Dose	Less tying gives a shorter delay at an identical parameter count (ρ = 0.86).
Proposal	**Encoder tying is one cause of grokking in conditional generative models.**

■ Appendix | Backup slides for questions

Appendix: index

A1 What is new vs. prior work?

A2 Does training itself differ?

A3 How were the thresholds chosen?

A4 Do the thresholds matter?

A5 What does the task teach?

A6 Do angles really appear?

A7 Video mod 23, as video

A8 Is it just more parameters?

A9 Real grokking, or just slow learning?

A10 What we did not test

A11 Do the two gradients conflict?

A12 Why one-step generation?

A13 How is a generated image scored?

A14 Is 16 runs enough? (all pairs, tests)

A15 Where does tying appear in practice?

A16 The 4 runs that break the pattern

A17 Training settings

A18 The XOR task

A1 What is new compared with prior work

New: the cause sits inside the conditioning mechanism

Work	Setting	What it shows
Power+, 2022	Transformers, classification	Grokking on algebraic tasks
Liu+, 2023	Classification, incl. images	Delay depends on weight norm / initialization
Kim+, 2026	Diffusion, modular images	Delay depends on data per class, patch size, block layout
This work	**Conditional flow matching**	**A cause unique to conditional generators (tying), with a dose-response**

New factor

Earlier factors are data, optimization, and blocks. Ours is how the condition enters.

New control

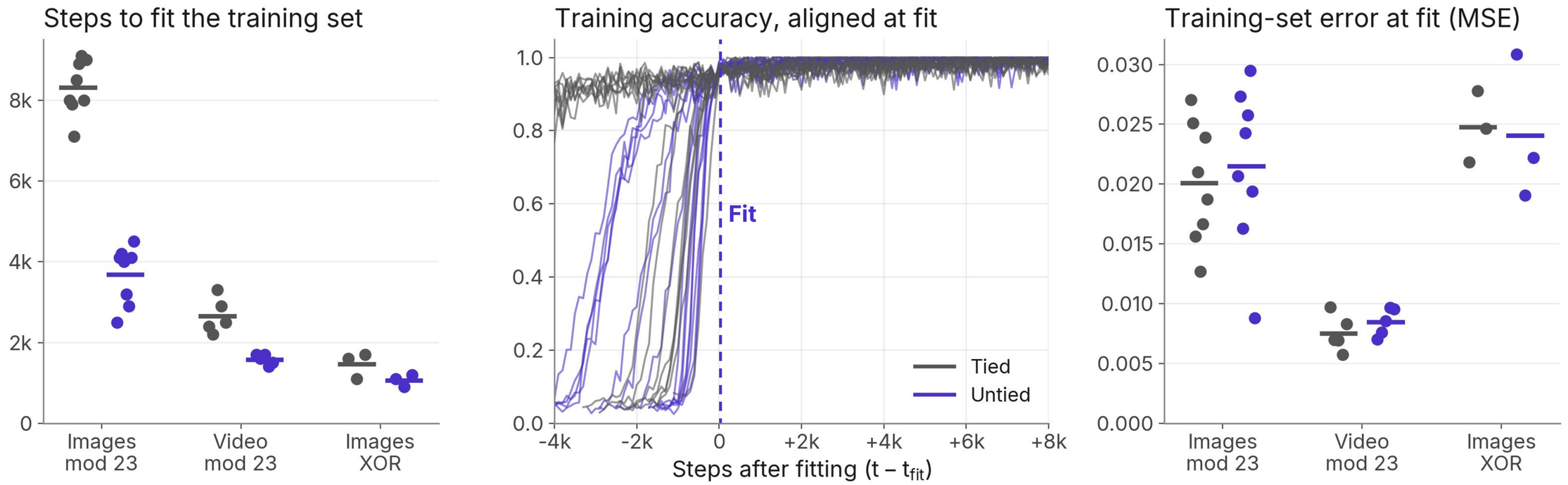
Tying is set to on, off, and in between, at an identical parameter count.

New scope

The same effect on images and video, and on mod 23 and XOR.

A2 Training differs in speed, not in the state at fit

Tied fits later, but both reach the same training state

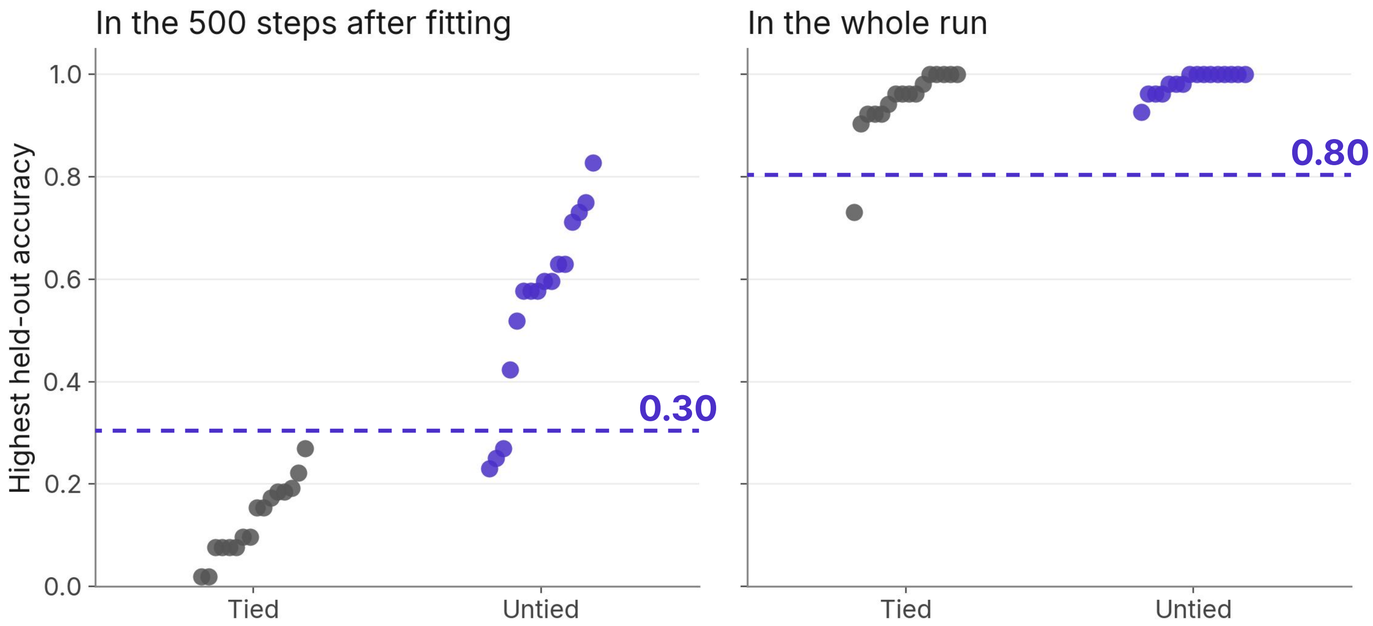


So we compare at the fit point: training state matches, held-out accuracy does not (A9)

Same training pairs, split, and optimizer in both. Mean MSE at fit (Tied vs Untied): images 0.020 vs 0.021, video 0.008 vs 0.008, XOR 0.025 vs 0.024. Mean steps to fit: images 8.3k vs 3.7k, video 2.7k vs 1.6k, XOR 1.5k vs 1.1k.

A3 How the thresholds were chosen

Fixed before the comparison, then stress-tested (A4)



Train ≥ 0.95

Fitted: at most 5% of training pairs wrong.

Held out < 0.30 for 500 steps

Clearly unsolved, yet above chance (0.04–0.06). 500 steps = 5 evaluations.

Held out ≥ 0.80

Clearly solved: 31 of the 32 runs reach it.

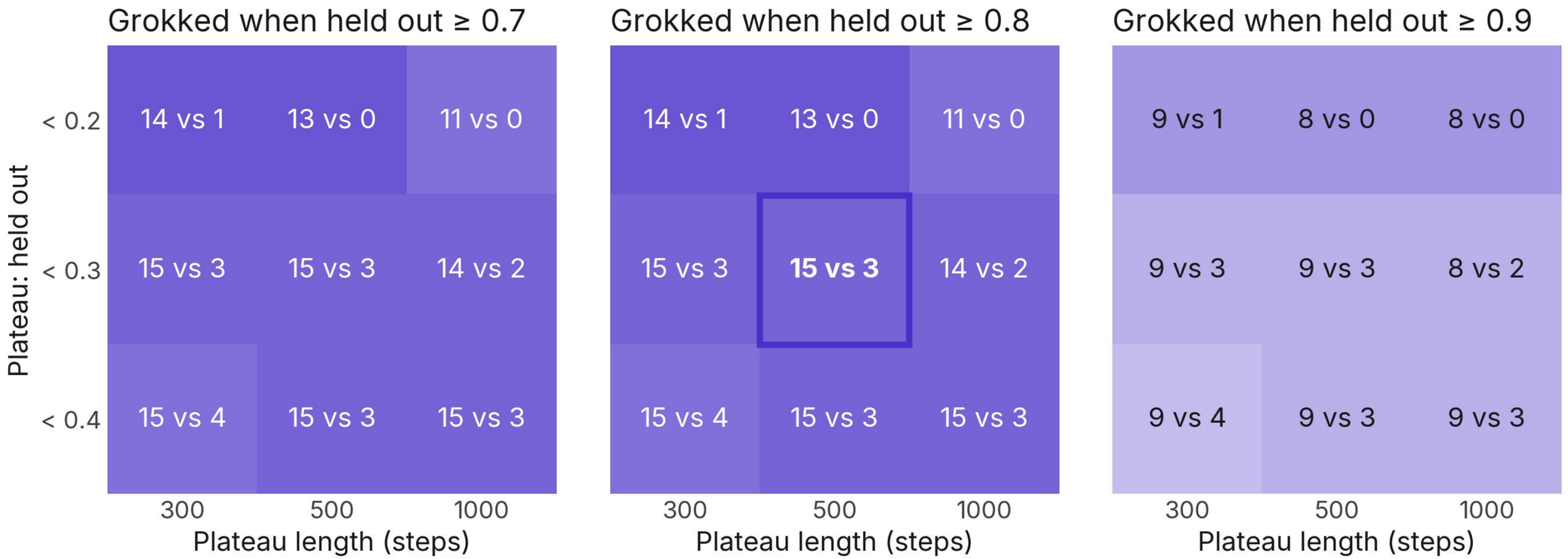
3 evaluations in a row

One lucky evaluation does not count.

We know of no standard numeric definition of grokking. Left: no Tied run exceeds 0.27; the 3 Untied runs below 0.30 are the 3 that meet the rule (A16). The rule was written before the first Tied vs Untied run.

A4 The result does not depend on the thresholds

Runs that meet the rule: Tied vs Untied (out of 16 each), for 27 threshold settings



Box = the rule used in the talk. Tied is always far ahead (8–15 vs 0–4). At ≥ 0.9, many Tied runs simply do not reach 0.9 within 30k steps. Accuracy just after fitting (slide 12) uses no threshold at all.

A5 What the task asks the model to learn

On a circle, mod 23 is simply adding angles

Is it prone to grokking?

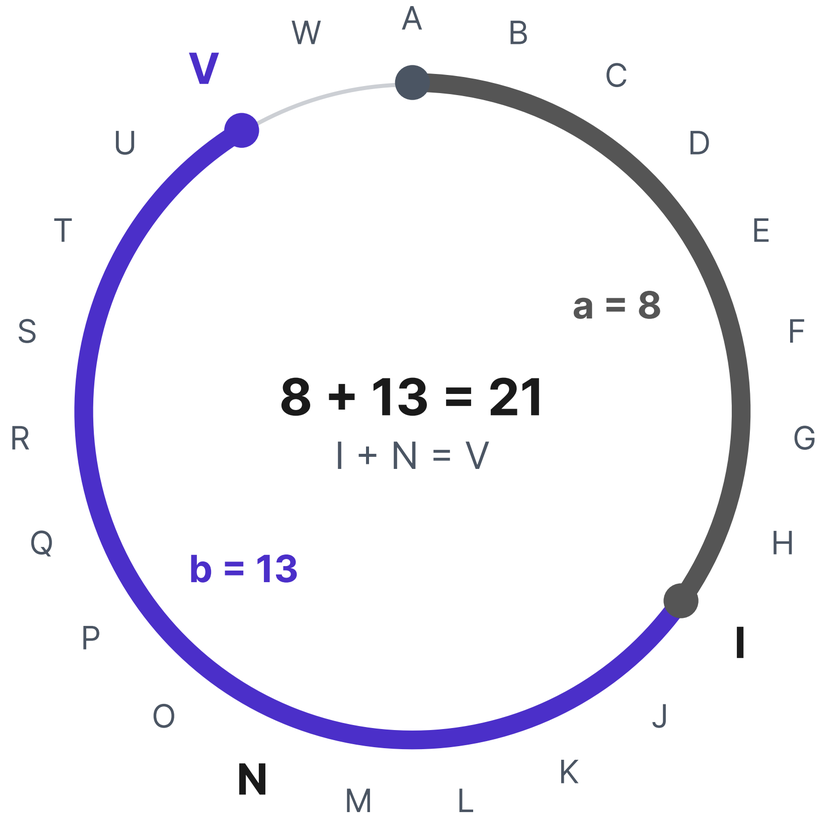
Yes: the standard grokking task (Power+, 2022). But the task alone is not enough: Tied 94% vs Untied 19% of runs.

How hard is it?

Letters carry no arithmetic, and the 28 held-out pairs cannot be memorized. Non-commutative groups generalize in neither model.

What must it acquire?

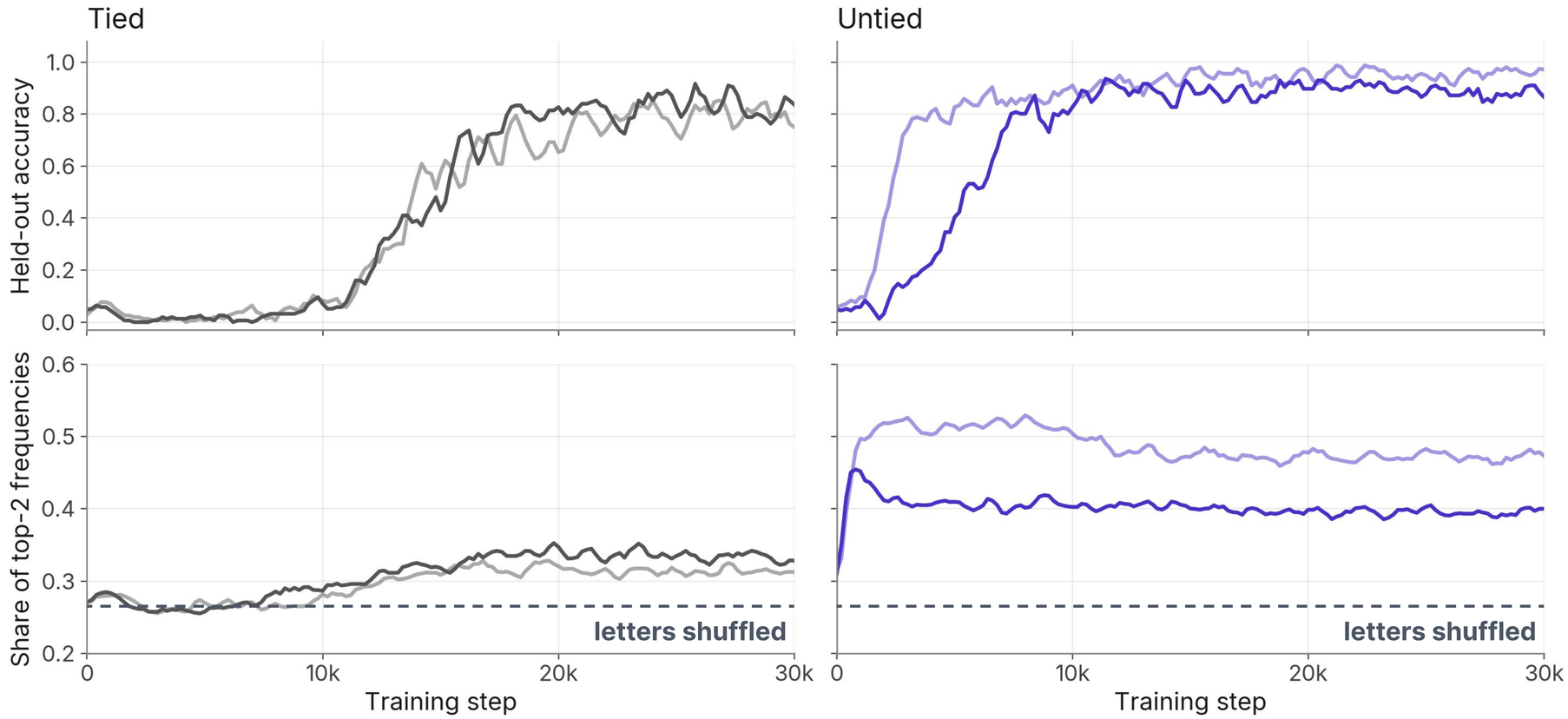
A rule that works on unseen pairs. In (x, y) coordinates the wrap-around at 23 has no simple rule; on a circle, adding is just turning.



Angle picture: the known solution for classifiers (Nanda+, 2023). Whether our generator uses it: first measurement on A6. It cannot be the whole story: XOR shows the same effect and has no circle (A18).

A6 Do the angles appear? A first measurement

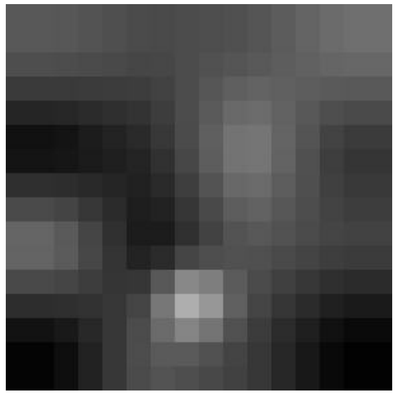
Periodic structure forms early in Untied, late in the plateau in Tied



Share of the letter-embedding variance in the 2 strongest of 11 Fourier frequencies (dashed: 95th percentile with letters shuffled, 0.26). Images mod 23; 2 seeds per arm, retrained with snapshots, same t_fit and t_grok as the original runs. Exploratory: consistent with an angle code, not a proof.

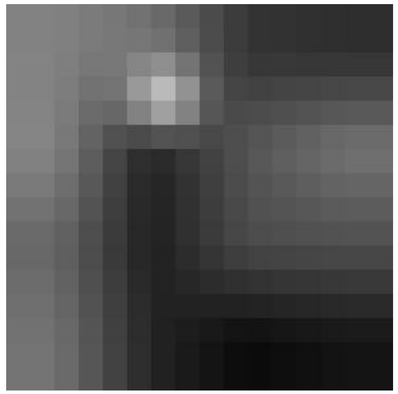
A7 Video mod 23, shown as video

A number is the angle of a moving blob: 6 + 16 = 22



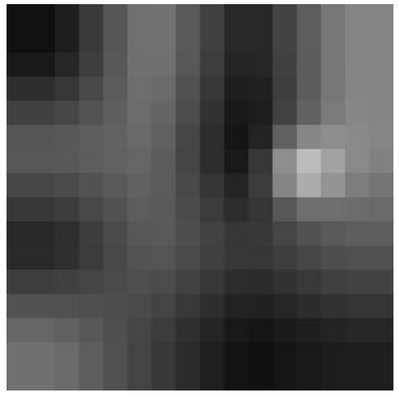
a = 6

+

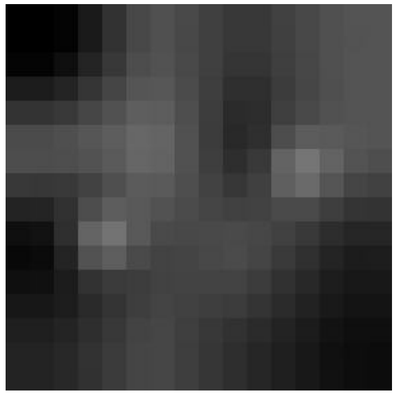


b = 16

=

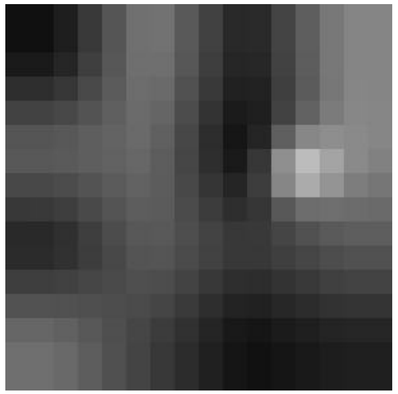


Correct answer: 22



On the plateau

step 4,000: nearest clip is 10



After grokking

step 30,000: nearest clip is 22

Number = angle

At frame 0 the blob sits at θ = 2πk / 23. It then circles once.

Output = sum of angles

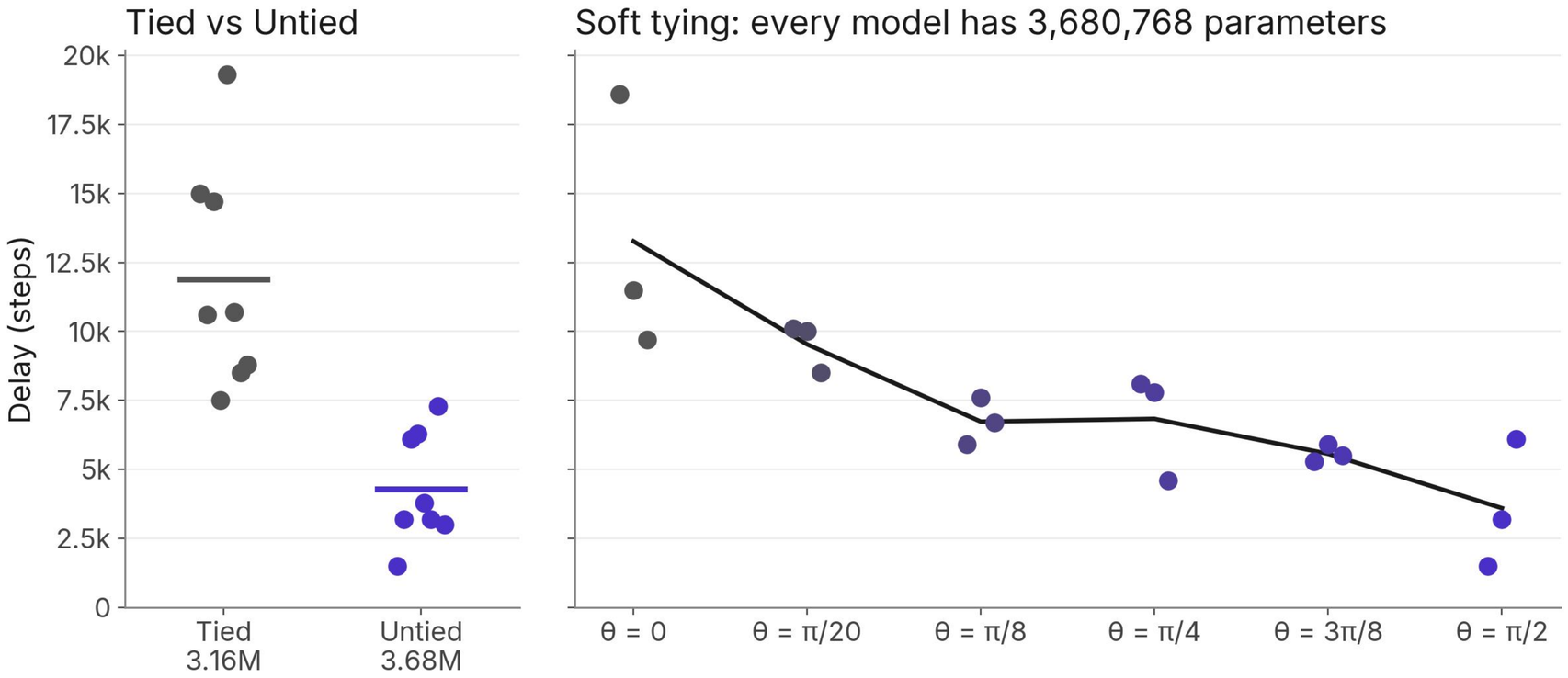
The model must generate the clip whose start angle is θ_a + θ_b.

Same held-out pair

The same Tied model, before and after grokking, on a pair it never trained on.

16×16 pixels, 8 frames, enlarged. Each clip also has a fixed background texture. Tied run, seed 0 (t_fit = 3,200, t_grok = 19,200); the model starts from pure noise.

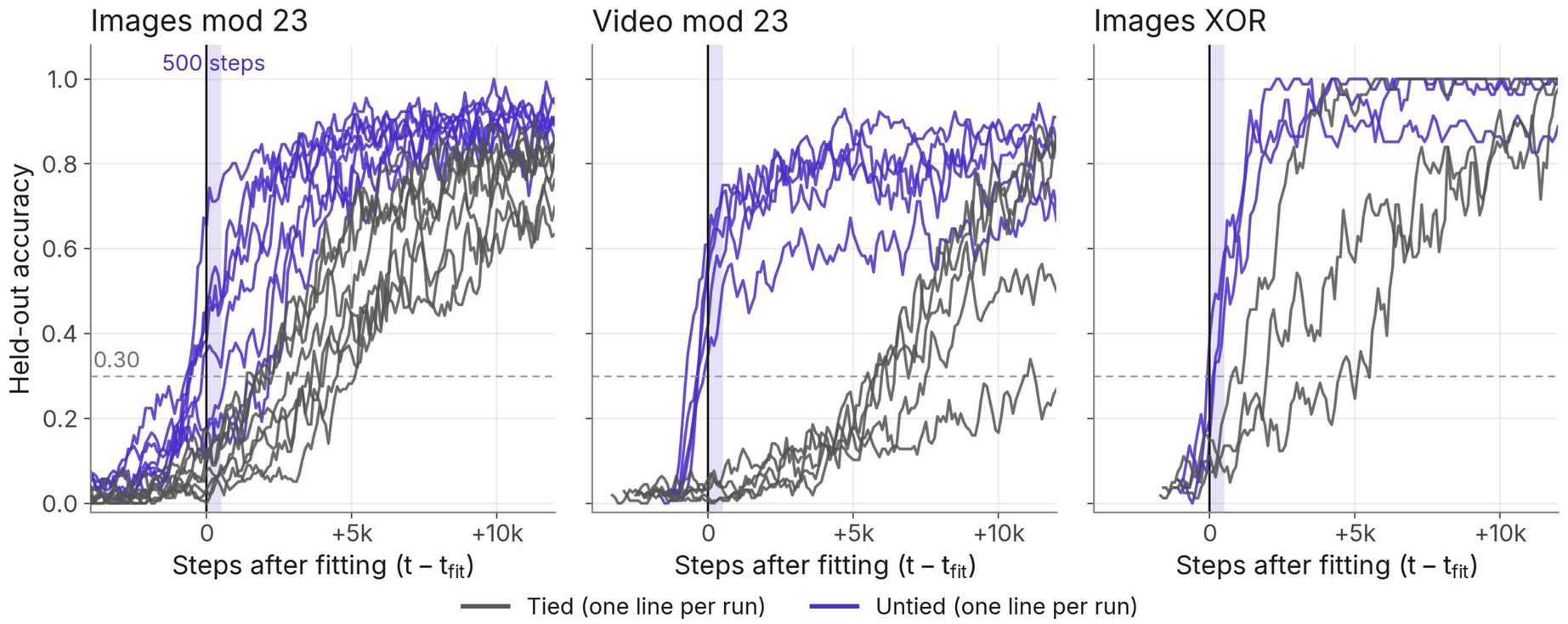
A8 Not capacity: same parameters, different delay



Untied has more parameters (+17% images, +25% video). Soft tying removes this difference, and the delay still changes from about 13k to 3.6k steps.

A9 Tied memorizes first; Untied does not

Every run, aligned at the moment it fits the training set

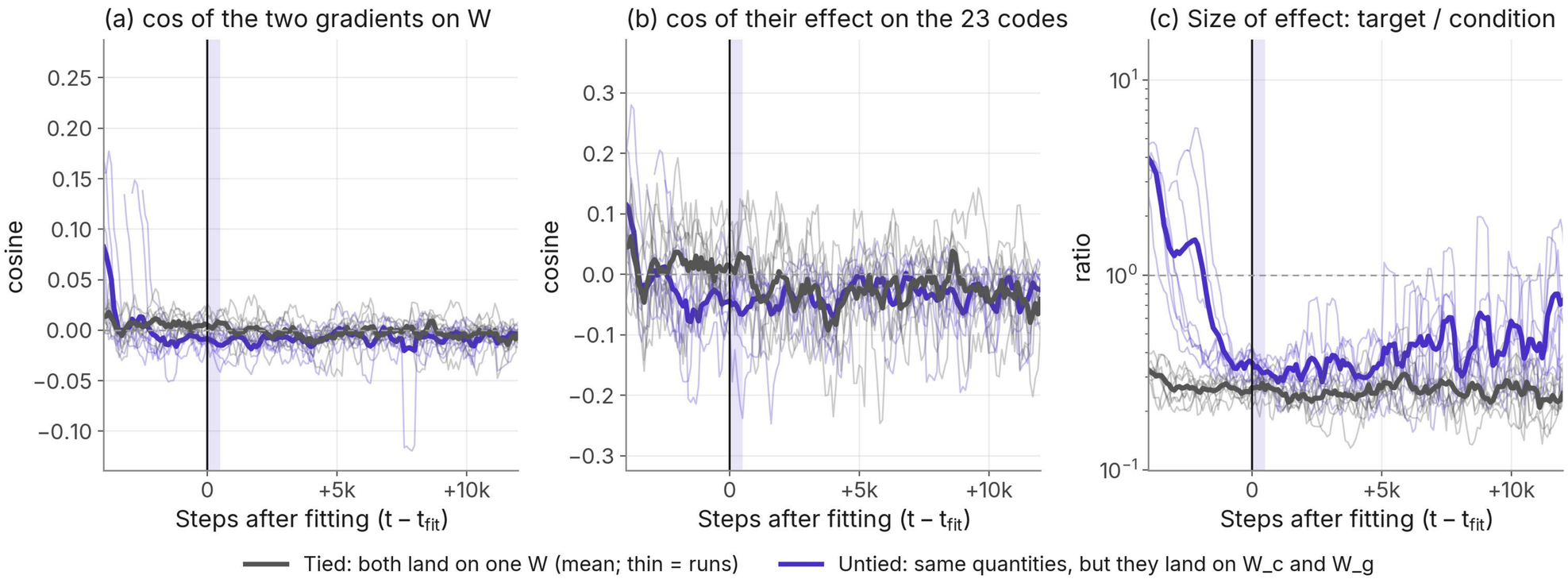


Shaded band: the 500 steps in which held-out accuracy must stay below 0.30 (dashed). Tied lines stay low after fitting; Untied lines are already rising.

A10 What we tested, and what we did not

Tested	Not yet tested
1-layer transformer, width 512	Deeper and larger models
Synthetic letter images and synthetic video	Natural images and real video
One-step generation	Many-step sampling
Addition mod 23 and XOR	Other conditional tasks
Tying of the input embedding	Tying in other layers

A11 Do the two gradients on W conflict?



No direct conflict: cos ≈ 0 in Tied and Untied alike (a, b).

Extra push: in Tied, the noisy-image gradient moves the condition codes by **~25%** of the main push (c).

8 + 8 runs (images mod 23), all training pairs, every 100 steps; t_fit and t_grok match the reported runs exactly. Raw gradients, before Adam's scaling. Untied values are only a reference (the parts land on W_c and W_g).

A12 Training and one-step generation

Training

Mix the true image c with noise:

$$x_t = (1 - t)\, c + t\, \varepsilon, \quad t \sim U(0, 1)$$

The model sees a, b, x_t, t and predicts c.

Loss: $\|f(a, b, x_t, t) - c\|^2$

Generation

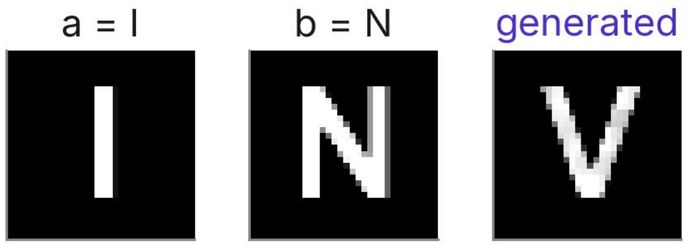
Start from pure noise, t = 1.

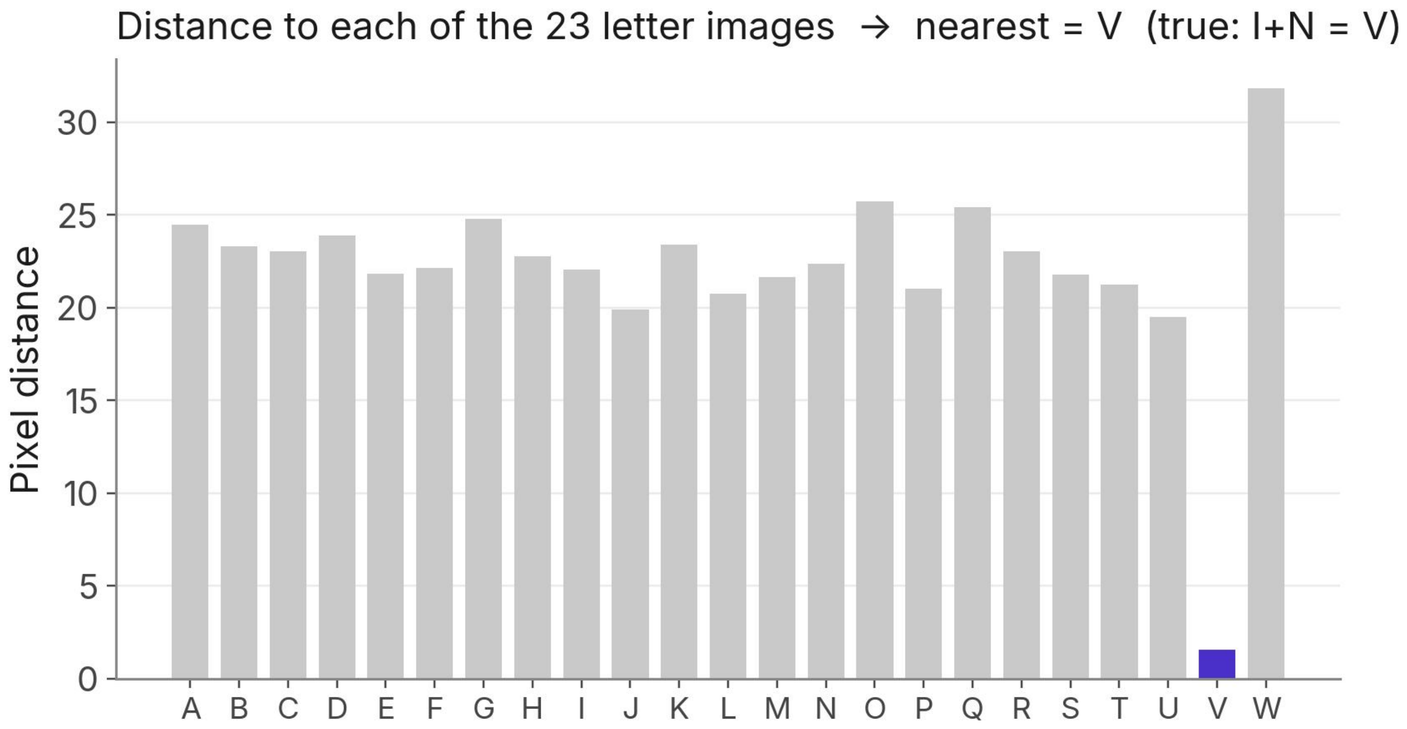
One forward pass gives the image c′.

→ Every evaluation is cheap, and the only part that differs between the two models is the embedding.

A13 Scoring: the nearest letter image

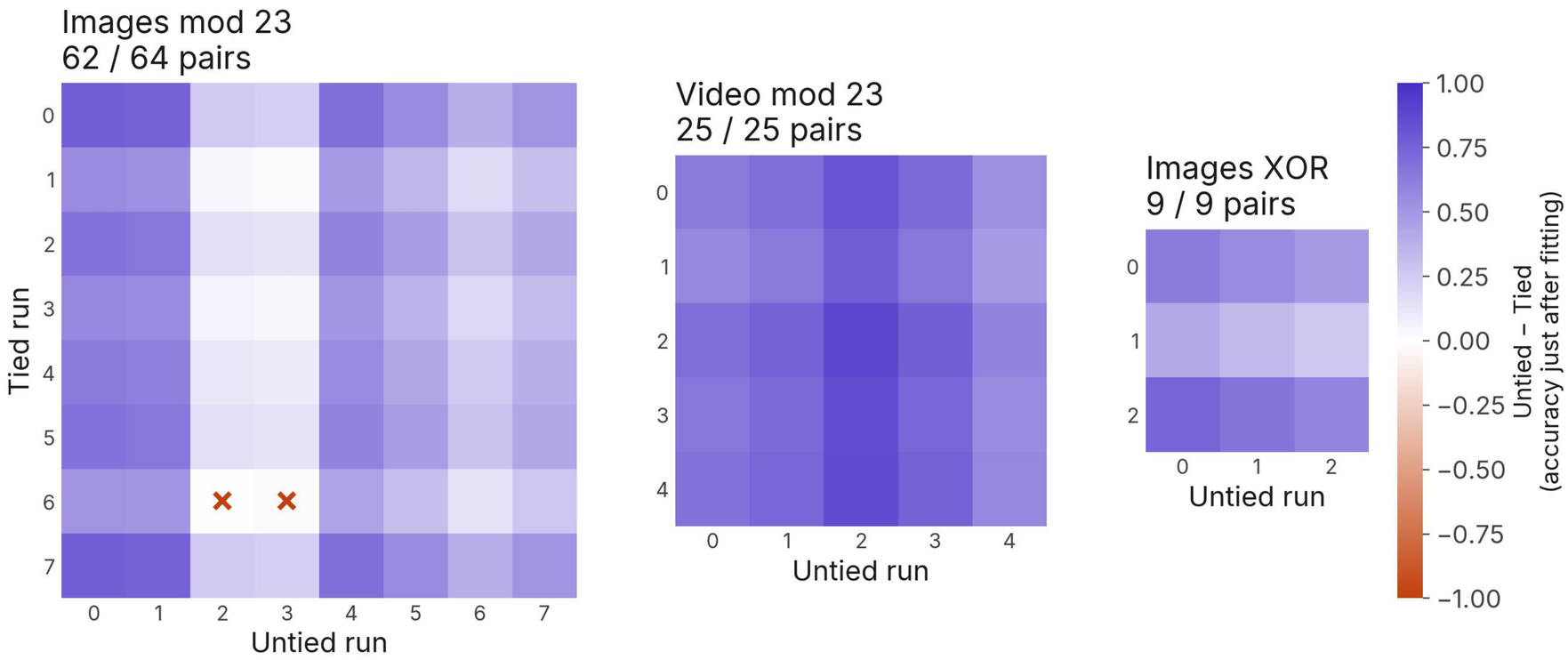
No trained classifier: compare with the 23 letter images and pick the nearest





A real held-out pair from the Tied run on slide 3, at step 30k. Accuracy = share of pairs whose nearest letter is the correct one.

A14 Every Tied–Untied pair, and a test

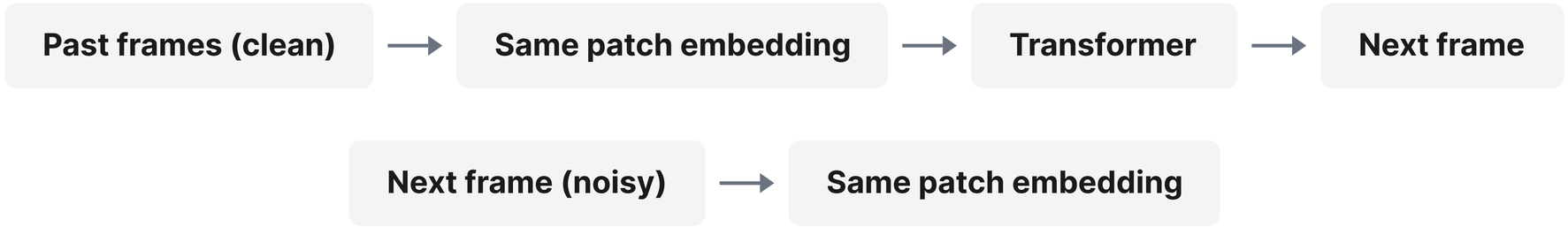


	Images mod 23	Video mod 23	Images XOR
Accuracy just after fitting	p = 0.0008	p = 0.004	p = 0.05
Delay	p = 0.0005	p = 0.03	p = 0.05

× = Untied not higher (1 tie, 1 loss). One-sided Mann–Whitney U; with 3 vs 3 runs, 0.05 is the smallest possible p.

A15 Where tying appears in practice

When the condition and the target are the same kind of data, one input layer for both is a natural choice



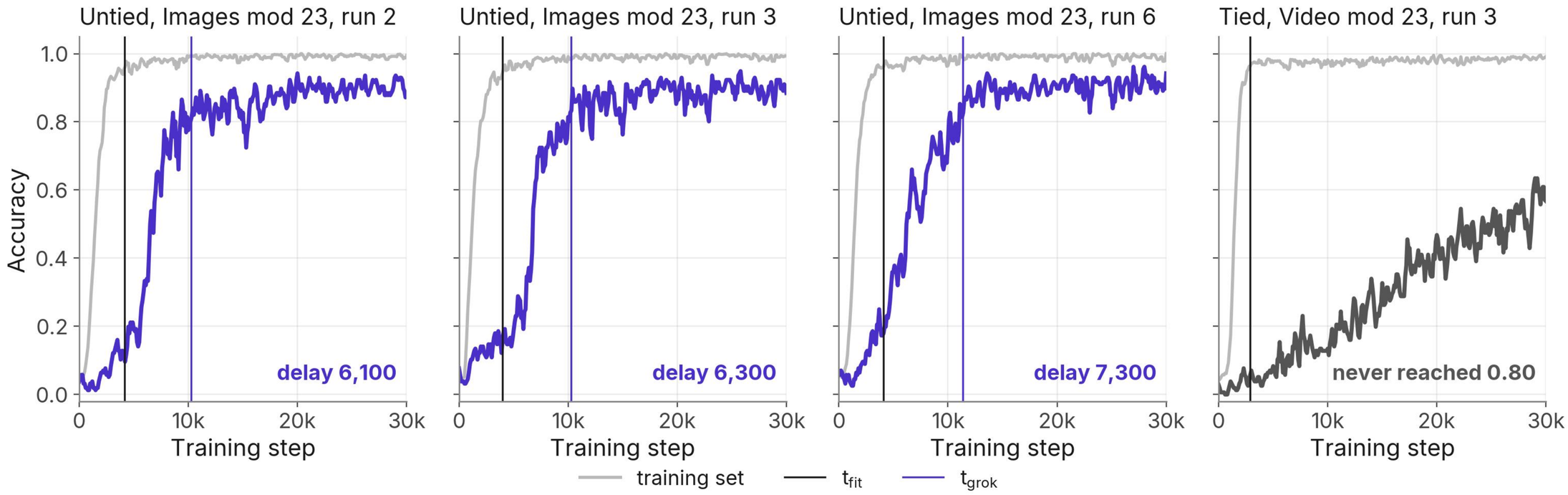
Why people tie

- Fewer parameters
- Simple to implement

Our suggestion

If a conditional model generalizes slowly, check whether conditions and the noisy target share an embedding, and try untying it.

A16 The four runs that break the pattern



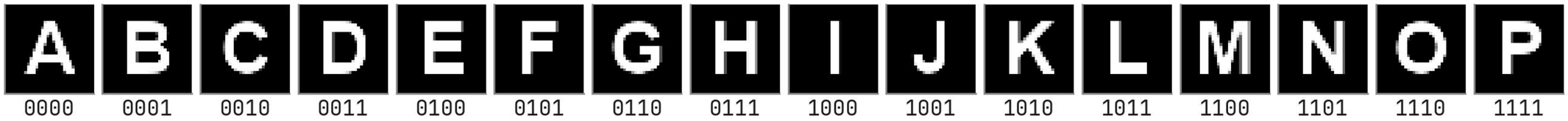
Three Untied image runs meet the rule, but their delays (6.1k–7.3k) are shorter than every Tied image run (7.5k–19.3k). The Tied video run still has a long low phase; it reaches only about 0.6 by step 30k.

A17 Training settings (identical for Tied and Untied)

	Images mod 23	Video mod 23	Images XOR
Data	32×32 letters	16×16 pixels × 8 frames	32×32 letters, 16 symbols
Pairs	276 (248 train / 28 held out)	276 (248 / 28)	136 (about 90% train)
Parameters: Tied → Untied	3.16M → 3.68M	4.21M → 5.25M	3.16M → 3.68M
Model	1-layer transformer, width 512, 16 heads	same	same
Optimizer	AdamW lr=0.001, weight decay=0.3, β=(0.9, 0.98)	same	same
Training	batch 64, 30,000 steps, evaluated every 100 steps	same	same
Runs per design	8	5	3

A18 The XOR task

16 symbols, each a 4-bit number; c = a XOR b



Example: **F** ⊕ **K** = **P**

0101 ⊕ 1010 = 1111

A different group from mod 23: **not cyclic**. Same images, model, and training settings.